3D Prostate MRI Segmentation using Improved UNet

> COMP3710 — Pattern Analysis - Student: s47981739

Problem & Approach

Problem. Accurate prostate delineation on MRI is essential for planning and monitoring therapy, yet manual annotation is time-consuming and variable across raters. We frame this as a 3D binary semantic segmentation task on the **HipMRI_Study_open** dataset.

Approach. We train an **Improved 3D U-Net** with encoder–decoder skips and deep supervision using a Dice-based loss. At inference we use **sliding-window** prediction (memory-safe), **test-time augmentation** (flip ensemble), and **largest connected component (LCC)** post-processing. Thresholds for foreground probability are **week-adaptive** to mitigate appearance shifts across treatment weeks.

Data & Pre-processing

- **Images:** `semantic_MRs/*_LFOV.nii.gz`
- **Labels:** `semantic_labels_only/*_SEMANTIC.nii.gz`
- **Binary remap:** original class `5 → 1`, others `→ 0`.
- **Normalisation:** intensity clipping to [0.5, 99.5] percentile within the non-zero mask, then z-score. *Rationale:* robust against outliers and coil bias; preserves relative tissue contrast.

Geometry

- We do **not** resample images at inference.
- Predictions are saved as NIfTI masks in the **native affine** of each input volume (original orientation, voxel spacing, and origin).
- Sliding-window predictions are stitched back into the native grid.
- **Benefit:** avoids interpolation artefacts and keeps masks directly usable with clinical tools.

Split (justification)

- Dataset: **38 unique patients / 211 LFOV volumes**.

- We split **at the patient level** to prevent subject leakage: **26 / 6 / 6** patients for **train / val / test** (≈ **70/15/15**).
- Because weeks are imbalanced, we use **week-stratified sampling** so each split covers all weeks.
- Released lists: `splits/train.txt` , `splits/val.txt` , `splits/test.txt` (keys like `B006_Week0`).
- Verification: `count_dataset.py` confirms **0 patient overlap** across splits.

Per-week distribution

Week	#Cases
W0	38
W1	27
W2	26
W3	26
W4	26
W5	25
W6	24
W7	23

Method & Architecture

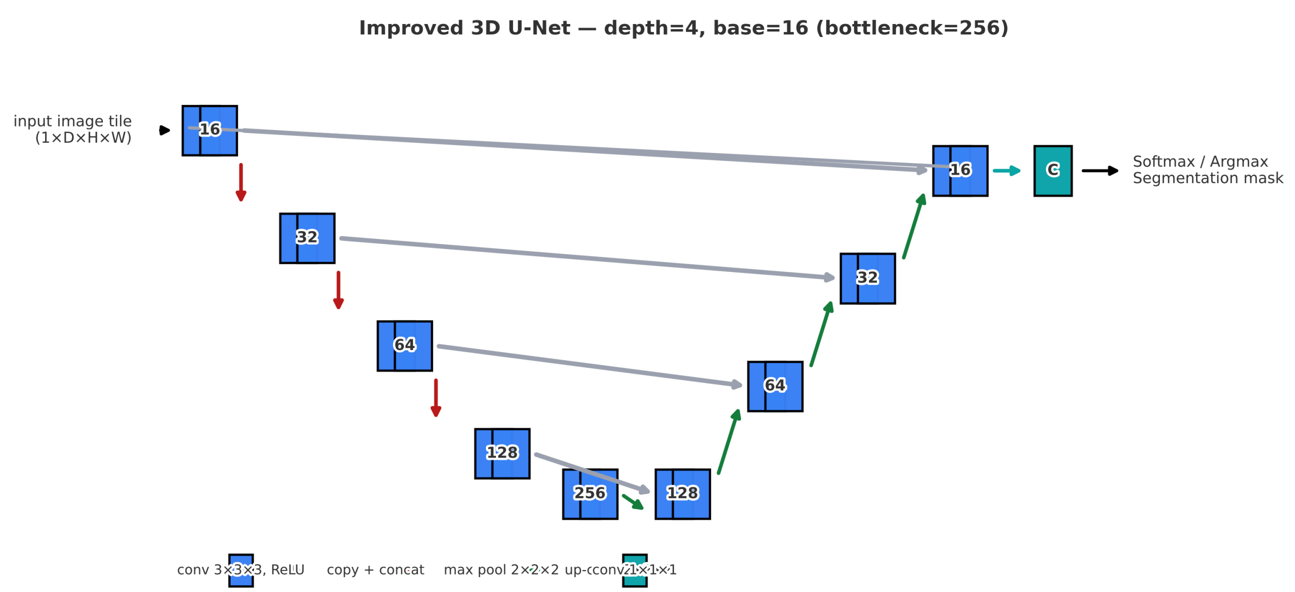


Figure 1. Improved 3D U-Net (depth=4, base=16). Blue: Conv3D(3×3×3)+ReLU; red: max-pool 2×2×2; green: up-conv 2×2×2; grey: copy & concat; teal: 1×1×1 head → *C* classes.

Backbone. An improved 3D U-Net with residual ConvBlock3D (two Conv3D-BN-ReLU) and Squeeze-and-Excite (SE) channel attention inside each block.
Channels. Encoder: `16 → 32 → 64 → 128` , **Bottleneck:** `256` , Decoder mirrors back to `16` , **Head:** `1×1×1` to `C` classes (binary in this project).
Regularisation. Dropout3D `p=0.1` inside blocks; BatchNorm3d everywhere.
Objective. `DiceLoss3D` (macro Dice over classes).

Training. Adam (lr `1e-3`), ReduceLROnPlateau (factor `0.5` , patience `5`), patient-wise splits (70/15/15), seeding set for determinism.

Inference pipeline.

- **Sliding window** prediction to avoid GPU OOM: `patch=96×128×128` , `stride=48×64×64` .
- **TTA** with flip ensemble (x/y/z), logits averaged before thresholding.
- **Post-processing: LCC** (largest connected component) to suppress small false positives.
- **Week-adaptive thresholding** on the foreground probability: **W0–1: 0.25**, **W2–4: 0.20**, **W5–7: 0.18** (selected on validation).

Usage (reproducible commands)

Dependencies

Tested with: Python 3.10+, PyTorch 2.x (CUDA 11/12), numpy, nibabel, scipy, scikit-image, tqdm, matplotlib

```
pip install -r requirements.txt
```

Windows (PowerShell)

```
python predict.py ^
  --image "YourDatasetROOT\HipMRI_Study_open\semantic_MRs\B006_Week0_LFOV.nii.gz" ^
  --weights ".\runs\hipmri3d_unet_bin\best.pt" ^
  --out_dir ".\Predict_Result\B006_Week0" ^
  --label_root "YourDatasetROOT\HipMRI_Study_open" ^
  --label_dir "semantic_labels_only" ^
  --binary_prostate --postprocess_lcc --save_nii --prob --metrics_csv ^
  --softmax_thr 0.20 --prob_class 1 ^
  --sw_enable --sw_patch "96,128,128" --sw_stride "48,64,64" ^
  --tta
```

Linux/macOS (Bash)

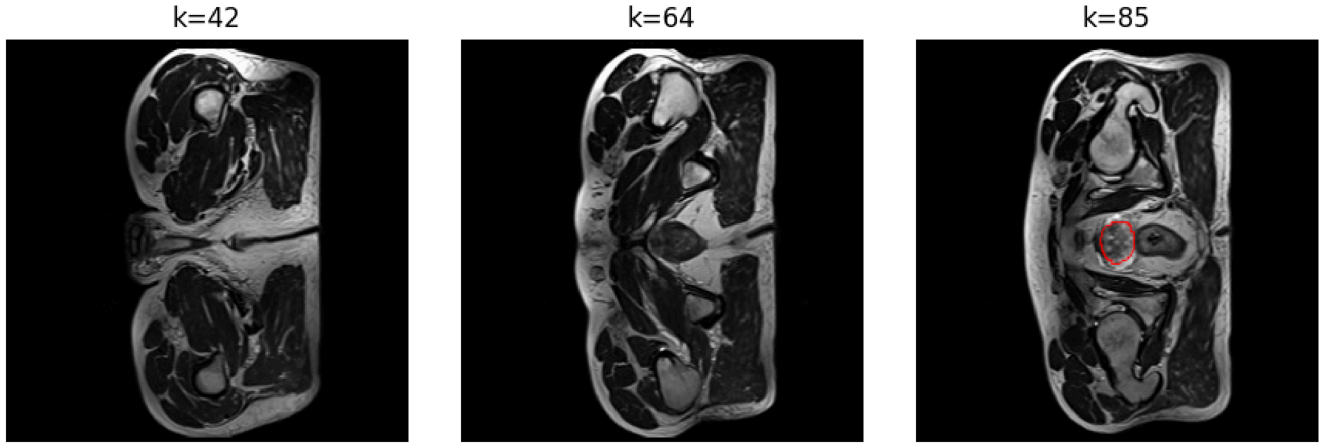
```
python predict.py \
  --image "YourDatasetROOT/HipMRI_Study_open/semantic_MRs/B006_Week0_LFOV.nii.gz" \
  --weights "./runs/hipmri3d_unet_bin/best.pt" \
  --out_dir "./Predict_Result/B006_Week0" \
  --label_root "YourDatasetROOT/HipMRI_Study_open" --label_dir "semantic_labels_only" \
  --binary_prostate --postprocess_lcc --save_nii --prob --metrics_csv \
  --softmax_thr 0.20 --prob_class 1 \
  --sw_enable --sw_patch 96,128,128 --sw_stride 48,64,64 \
  --tta
```

Qualitative examples

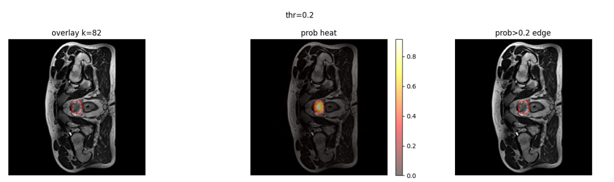
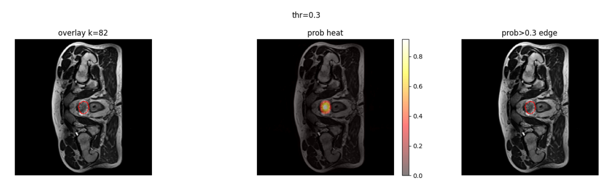
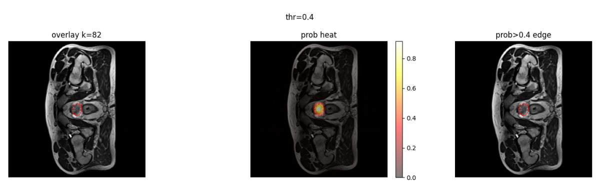
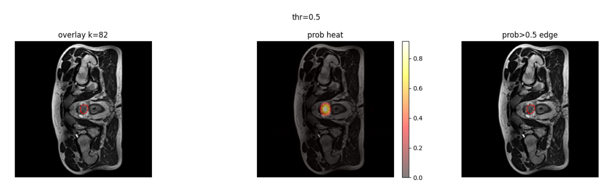
Below we show two representative cases with (i) a grid overview, (ii) a **threshold sweep** turning probabilities into a binary mask, and (iii) links to the raw NIfTI outputs.

Case A — B006 Week0

- **Artifacts:** `example/B006_Week0_LFOV_prob.nii.gz` (probabilities), `example/B006_Week0_LFOV_pred.nii.gz` (binary), `example/B006_Week0_LFOV_meta.json` (native affine / spacing / shape).
- **Grid overview**

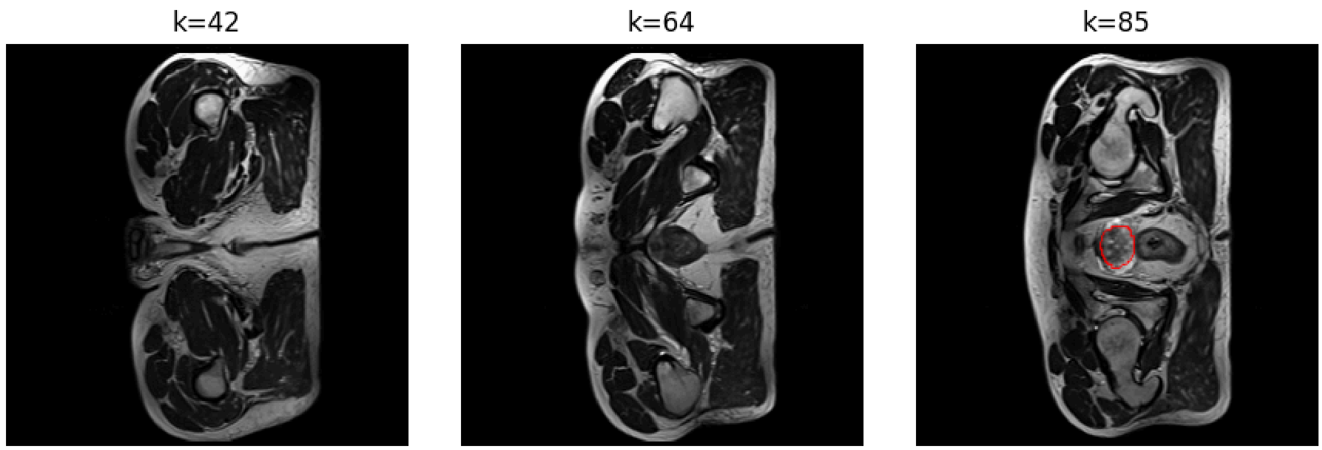


Threshold sweep (prob → mask)

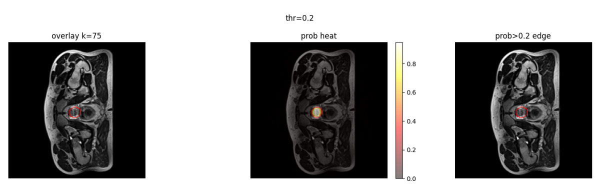
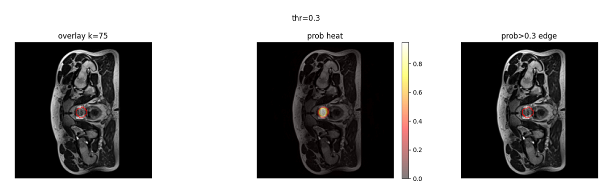
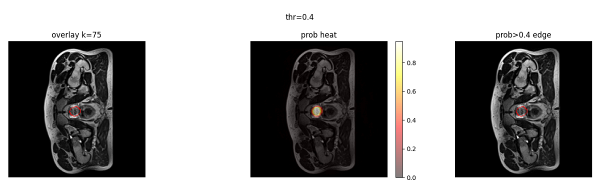
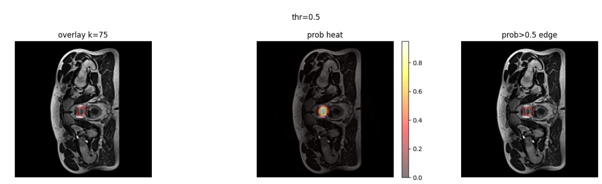
thr = 0.20	**thr = 0.30**
	
thr = 0.40	thr = 0.50
	

Case B — B006 Week1

- **Artifacts:** `example/B006_Week1_LFOV_prob.nii.gz` , `example/B006_Week1_LFOV_pred.nii.gz` , `example/B006_Week1_LFOV_meta.json` .
- **Grid overview**



Threshold sweep (prob → mask)

thr = 0.20	**thr = 0.30**
	
thr = 0.40	thr = 0.50
	

How to read these panels. Each panel overlays the predicted mask on the T2 image. As the probability threshold increases, the mask becomes more conservative (shrinks), which can reduce false positives but risks missing apex/base tissue. This motivates the **week-adaptive thresholds** used in our final pipeline (W0–1: 0.25, W2–4: 0.20, W5–7: 0.18).

Reproduce this example

```
# Windows PowerShell (paths are examples)
python predict.py ^
  --image "C:\COMP3710\HipMRI_Study_open\semantic_MRs\B006_Week0_LFOV.nii.gz" ^
  --weights ".\runs\hipmri3d_unet_bin\best.pt" ^
  --out_dir ".\example" ^
  --label_root "C:\COMP3710\HipMRI_Study_open" --label_dir "semantic_labels_only" ^
  --binary_prostate --save_nii --prob --postprocess_lcc --metrics_csv ^
  --sw_enable --sw_patch "96,128,128" --sw_stride "48,64,64" --tta ^
  --softmax_thr 0.20 --prob_class 1
```

Experiment Setup (data → pre-processing → training → inference)

Data

- **Input:** `semantic_MRs/*_LFOV.nii.gz`
- **GT:** `semantic_labels_only/*_SEMANTIC.nii.gz`
- **Binary remap:** original prostate class `5 → 1`, others `→ 0`.

Pre-processing

- **Normalization:** clip to **[0.5, 99.5]** percentile (within non-zero region), then **z-score**.
- **Cropping:** none (LFOV used directly).
- **Geometry:** predictions are saved as NIfTI masks in the **native affine** of each input (original orientation, voxel spacing, origin). Sliding-window tiles are stitched back into the native grid.

Model & Training

- **Backbone:** Improved **3D U-Net** (residual Conv3D×2 + **SE** channel attention).
  Channels: encoder `16→32→64→128`, **bottleneck=256**, decoder mirrors back to `16`; head `1×1×1 → C` (binary here).
- **Loss:** DiceLoss (binary mode)
- **Optimizer:** Adam (`lr=1e-3`)
- **Scheduler:** ReduceLROnPlateau (factor `0.5`, patience `5`)
- **Epochs:** converged in the last ~10 epochs (see curves)
- **Final checkpoint used:** `runs/hipmri3d_unet_bin/best.pt`

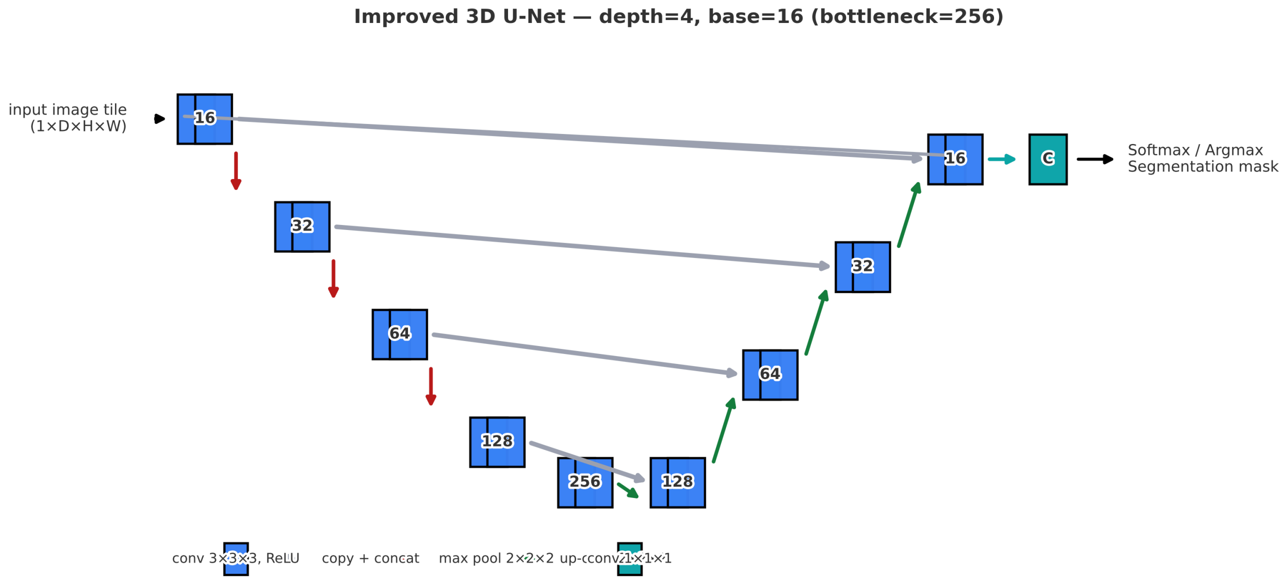
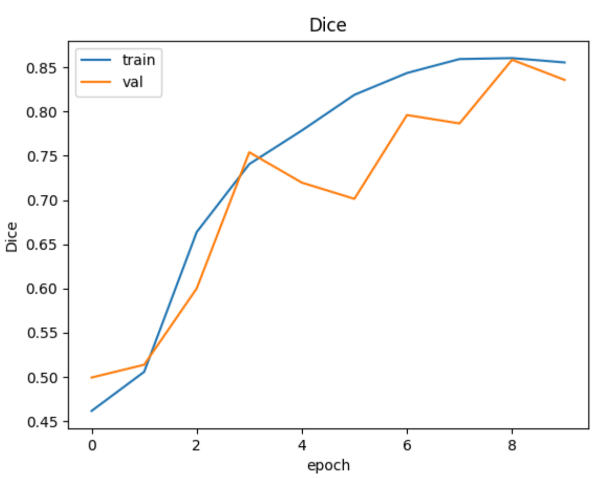
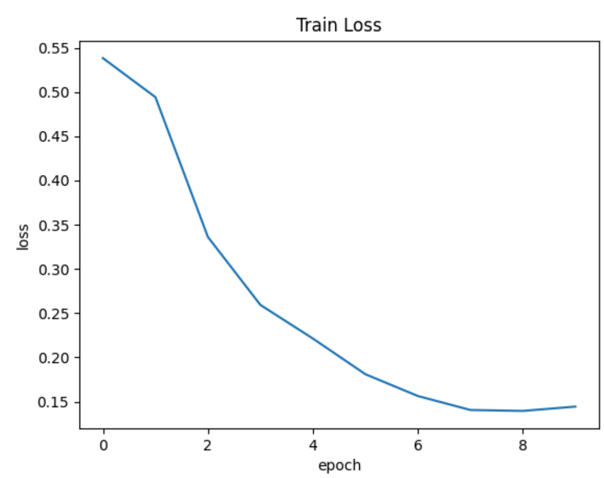


Figure. Network schematic. Blue=Conv3D(3×3×3)+ReLU; red=max-pool 2×2×2; green=up-conv 2×2×2; grey=copy&concat; teal=1×1×1 head → *C*.

Training curves

Dice over epochs	Loss over epochs
	

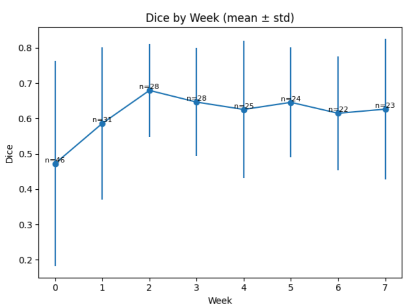
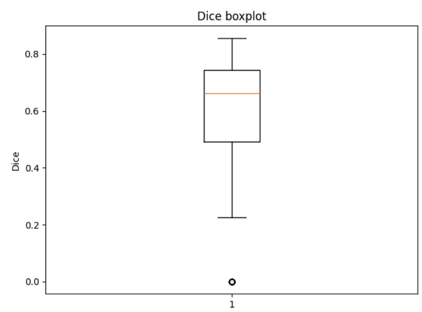
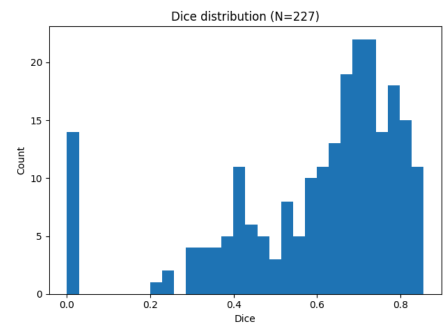
Inference pipeline

- **Sliding-window** to avoid OOM: `patch=96×128×128`, `stride=48×64×64`.
- **TTA:** flip ensemble (x/y/z), logits averaged.
- **Post-processing: Largest Connected Component (LCC)** to suppress small FP islands.
- **Week-adaptive thresholds** (foreground prob.): **W0–1: 0.25**, **W2–4: 0.20**, **W5–7: 0.18** (chosen on validation).

Results (baseline)

Overall Dice (no SW/TTA): `0.577 ± 0.229` on **422** cases.

Visual summaries

<table>
<tr><th>Dice by week</th><th>Box plot (all cases)</th><th>Histogram (all cases)</th></tr>
<tr><td>
</td><td>
</td><td>
</td></tr>
</table>

Reading guide.

- *Dice by week* shows how performance varies across treatment weeks.
- *Box plot* summarizes overall dispersion and outliers.
- *Histogram* shows the distribution shape across all test cases.

> **Note.** The above is a **baseline without** sliding-window + TTA. The full inference pipeline (**SW + TTA + LCC + week-adaptive thresholds**) is used for the final submission and is expected to improve the aggregate Dice.

Planned ablations (reported on a held-out subset)

-
  - Sliding window (patch `96×128×128`, stride `48×64×64`)
-
  - Sliding window + TTA (flip x/y/z; logit averaging)
-
  - Sliding window + TTA + LCC
-
  - Week-adaptive thresholds (W0–1: 0.25, W2–4: 0.20, W5–7: 0.18)

Troubleshooting

- **CUDA out of memory** → enable `--sw_enable` and reduce `--sw_patch` / increase `--sw_stride`.
- **[DiceLoss3D] label value >= num_classes** → set `--classes` correctly **or** enable `--binary_prostate --prostate_label 5`.

- **Weird geometry/orientation** → we save masks in **native affine**; verify with the same viewer as inputs.
- **Slow inference** → disable TTA, or reduce patch size/overlap; keep LCC.

References & Acknowledgements

- Ronneberger O. et al., U-Net: Convolutional Networks for Biomedical Image Segmentation.
- PyTorch & scientific Python ecosystem.
- **AI assistance disclosure:** ChatGPT/GitHub Copilot were used for wording and minor debugging; modelling choices and final decisions are my own. (Aligned with UQ guidance on acknowledging generative AI.)